

Taller: Control Robusto y Estocástico

Tema: Modelando las incertidumbres

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Use the `ureal` uncertain element to represent real numbers whose values are uncertain when modeling dynamic systems with uncertainty. An uncertain real parameter has a nominal value, stored in the `NominalValue` property, and an uncertainty, which is the potential deviation from the nominal value. `ureal` stores this deviation equivalently in three different properties: `PlusMinus`, `Range` and `Percentage`.

```
clc, clear all, close all

delta1 = ureal('delta1',2);
delta2 = ureal('delta2',2,'Percentage',[-10 20]);

get(delta1)
get(delta2)
```

You can create an uncertain real parameter whose value can vary from 14 to 19, with a nominal value of 15.5.

```
p1 = ureal('p1',15.5,'Range',[14,19])
get(p1)
```

Create an uncertain real parameter with a nominal value of 24, whose value can increase or decrease by 15%.

```
k1 = ureal('k1',24,'Percentage',15)
get(k1)
```

Use the following to create a `ureal` parameter and an uncertain dynamic system model.

```
prm_1 = ureal('w0',10);
sys_1 = tf(prm_1,[1 prm_1])
sys_1.NominalValue
```

You could make a simple plot of your parameters

```
pmin = prm_1.Range(1);
pmax = prm_1.Range(2);
pnom = prm_1.NominalValue;
```

```

figure
plot([pmin pmax],[1 1], 'k-', 'LineWidth',2)
hold on
plot(pnom,1, 'bo', 'MarkerSize',12, 'LineWidth',3)

xlim([pmin-1 pmax+1])
ylim([0.5 1.5])
yticks([])
grid on

xlabel('Value of Parameter 1')
title('Uncertainty interval')
legend('Interval', 'Nominal')

```

Plot multiple uncertain parameters

```

delta1 = ureal('delta1',2);
delta2 = ureal('delta2',2, 'Percentage', [-10 20]);
p1 = ureal('p1',15.5, 'Range', [14 19]);
k1 = ureal('k1',24, 'Percentage',15);

parameters = {delta1,delta2,p1,k1};
names = {'delta1','delta2','p1','k1'};

figure
hold on, grid on

for i = 1:length(parameters)
    parameter = parameters{i};
    pmin = parameter.Range(1);
    pmax = parameter.Range(2);
    pnom = parameter.NominalValue;
    plot([pmin pmax],[i i], 'k-', 'LineWidth',2)
    plot(pnom,i, 'bo', 'MarkerSize',8, 'LineWidth',3)
end

yticks(1:length(parameters))
yticklabels(names)

xlabel('Parameter Value')
title('Intervals')
legend('Interval', 'Nominal')

```

Exercise 1.

Create a model of a second-order system with natural frequency $w_0 = 10 \pm 3$ rad/s and a damping ratio that can vary from 0.5 to 0.8, with a nominal value of $\zeta = 0.6$. First, represent the natural frequency and damping ratio values as uncertain real parameters.

```

% Parameters
w0 = ureal('w0',10,'PlusMinus',[-3 3]);
zeta = ureal('zeta',0.6,'Range',[0.1 0.9]);

% System
sys1 = tf(1,[1/w0^2 2*zeta/w0 1])
figure
impz(sys1, 'm--', sys1.NominalValue, 'r-')
legend('Uncertain', 'Nominal')

```

Exercise 2.

Create a state-space model with matrices having uncertain elements. **Tip:** you could first use uncertain parameters such as `ureal` and `ucomplex` and then use these elements to specify the state-space matrices of the system.

```

p = ureal('p',1);

A = [0 3*p; -p p^2];
B = [0; p];
C = ones(2);
D = zeros(2,1);
usys = ss(A,B,C,D)
usys.Nominal

```

Exercise 3.

Consider an RC circuit with transfer function $G(s) = \frac{1}{RCs + 1}$ with $C = 100\mu F$ and $R = 1K\Omega$. Find the step response when the Resistor and Capacitor tolerance varies 5%, 10% and 15%.

```

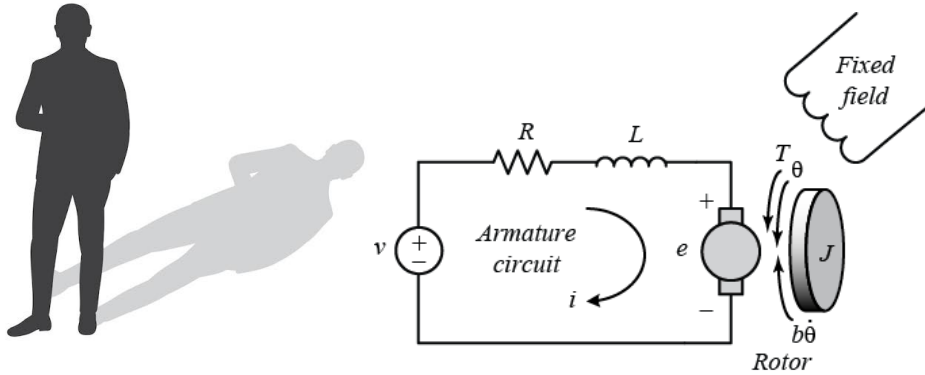
R = ureal('R',1000,'Percentage',15);
C = 100e-6;
%C = ureal('C',100e-6,'Percentage',10);
G = tf(1,[R*C 1]);

figure
step(G,G.NominalValue)
grid on
legend('Uncertain','Nominal')
title('Step Response')

```

The `ultidyn` objects represent uncertain linear time-invariant dynamics whose only known attributes are bounds on their frequency response. Combine `ultidyn` objects with other dynamic system models and uncertain elements to model systems with uncertain dynamics, represented by `uss` models.

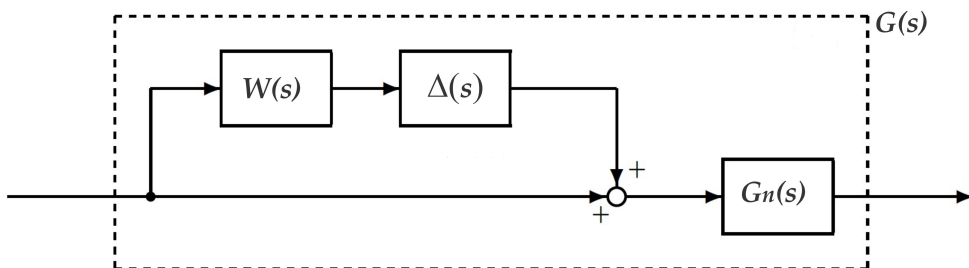
Shadow-Men Analogy



Suppose you want to make the transfer function identification of a dc motor and you find the nominal model

$$G_n(s) = \frac{5}{0.2s + 1}.$$

This model describes correctly the model behavior in low-frequencies. However, the motor has elasticity in the axis, mechanical vibrations, structural resonances and other unknown dynamics. Only we know is that its effect is limited. In this case the real plant can be represented as $G(s) = G_n(s)(1 + W(s)\Delta(s))$



where $W(s)$ is the weighting function and $\Delta(s)$ is the unknown dynamics with $\|\Delta_\infty\| \leq 1$.

Exercise 4.

Create an uncertain dynamic system model representing an uncertainty with gain of 0.1 at low frequencies and gain of 10 at high frequencies.

```
H = ultidyn("H",1) %Creation of a SISO ultidyn block with gain less
                    than 1 at all frequencies.
W = tf([1 0.1],[0.1 1]) %Creation a weighting function with gain starting a
                        0.1 and increasing to 10.
bodemag(W,{1e-2,1e2}), grid on
bodemag(W,'ro',W*H,'m-'), grid on, ylim([-60 30])
```

You can combine Delta with other models to introduce the dynamic uncertainty to your system. For example to model a system as a state-space

```
H2 = ultidyn("H",2)
Delta = W*H2
A = [-5 10;-10 -5];
B = [1 0;0 1];
C = [1 10;-10 1];
D = 0;
Gnom = ss(A,B,C,D);

Gadd = Gnom + Delta;           %Here the uncertainty is additive.
%Gmult = Gnom*(1+Delta);      %Here the uncertainty is multiplicative.
step(Gadd,5), grid on, ylim([-5 5])
```